

RESEARCH ARTICLE

Endurance training attenuates acute exercise-induced monocyte transcriptomic responses in adolescents: sex-specific molecular adaptations

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Abstract

Circulating monocytes contribute to atherogenesis and vascular dysfunction. Although clinical cardiovascular disease presents in adulthood, its biological origins often begin in youth and differ substantially between females and males. Twelve males and nine females (13–17 yr) completed an acute exercise protocol consisting of 10, 2-min cycling bouts at 70% of maximal work rate interspersed with 1-min rest intervals, performed before and after an 8-wk supervised endurance training intervention with brief supplementary strength work (60 min/session, 3 sessions/wk). Blood was collected before and immediately after each exercise challenge. Peripheral blood monocytes were isolated, and whole transcriptome RNA sequencing (RNA-seq) was performed. Before training, acute exercise induced a markedly greater monocyte transcriptomic response in females compared with males [5,135 vs. 567 differentially expressed transcripts, false discovery rate (FDR) < 0.1]. Pathway analyses identified vascular function-related pathways in males, whereas females showed enrichment of pathways related to adipose tissue cross talk and oxidative metabolism. Following training, the acute transcriptomic response was markedly attenuated in both sexes (165 transcripts in males and 94 in females, FDR < 0.1), representing an ~98% reduction in females and a ~70% reduction in males relative to pre-training responses. These findings reveal sex-specific monocyte responses to acute exercise in youth and suggest that endurance exercise alters immune transcriptional responsiveness in pathways relevant to vascular and cardiovascular health.

NEW & NOTEWORTHY Acute exercise induced a markedly greater monocyte transcriptomic response in female adolescents than in males. Females showed activation of pathways related to adipose tissue signaling and oxidative metabolism, whereas males exhibited vascular-function pathways. Following exercise training, the monocyte transcriptomic response to acute exercise was substantially attenuated in both sexes. These findings identify sex-specific immune transcriptional responses to exercise during adolescence with potential implications for cardiovascular health.

cardiovascular disease; gene expression; immune cells; pediatrics; physical activity

INTRODUCTION

Cardiovascular disease (CVD) is the leading cause of mortality worldwide, and its biological origins often begin during childhood and adolescence, long before the appearance of clinical symptoms (1). Monocytes, key innate immune cells involved in inflammation and vascular remodeling, play a central role in the early development of atherosclerosis (1). Monocyte gene expression patterns in youth have been linked to future cardiometabolic risk (2). Because exercise modulates immune function (3, 4), understanding how acute exercise and exercise training shape monocyte biology during this sensitive developmental window is important for early CVD prevention.

Genomic studies demonstrate that exercise alters monocyte gene expression and noncoding RNAs (ncRNAs) that regulate inflammatory and metabolic pathways implicated in atherosclerosis (5, 6), yet these responses are poorly characterized in

pediatric populations. Physiologic and genomic responses to exercise across development are sex-specific (7–10), and sexual dimorphism of monocyte gene expression profile has been reported in adults in the context of low-grade chronic inflammation (11). In earlier studies focused on gene expression of peripheral blood mononuclear cells (PBMCs), we found a remarkable 10-fold greater impact of brief exercise on PBMC gene expression in early-pubertal females compared with males (9, 10). Whether males and females differ specifically in monocyte transcriptomic responses to acute exercise, and whether training alters those responses, remains unknown.

Although monocytes play a key role in atherosclerosis in both women and men, growing evidence demonstrates substantial sexual dimorphism in the pathology and clinical manifestations of atherosclerosis (12–14). Women are more prone to fibrous plaques with endothelial dysfunction, coronary microvascular disease, and smooth muscle cell phenotypic switching



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(13, 14). Men are more likely to have obstructive coronary artery disease with larger lipid-rich necrotic cores, thinner fibrous caps, and greater inflammatory responses, resulting in increased plaque vulnerability at younger ages (15). Coronary heart disease is the most common first manifestation of CVD in men at every age, whereas women are more likely to present with stroke or heart failure as their first event, particularly after age 70 (15). Women also have unique risk factors, including pregnancy complications (e.g., preeclampsia and gestational diabetes), premature menopause, and exogenous hormone use (12). These observations highlight the imperative to explore sexual dimorphisms in the monocyte response to acute exercise and exercise training as early in life as possible.

We therefore aimed to characterize monocyte transcriptomic responses to an acute exercise bout in healthy adolescent females and males and to determine how these responses change after an 8-wk exercise training intervention. We designed our experiments to focus on endurance-training-associated genomic changes in monocytes following a brief bout of exercise. Increasing evidence from this and other laboratories indicates that immune cell adaptations to exercise training are more pronounced following an acute exercise bout than in cells obtained at rest (16, 17). Since the acute physiological and immunogenomic responses to exercise are highly dependent on the relative magnitude of the brief exercise perturbation (18), we scaled the acute work rate input to each subject's physical fitness level.

Building on our previous findings in young adult males (5) where acute exercise shifted monocytes toward an anti-inflammatory, antiatherogenic program (e.g., downregulation of *TNF*, *TLR4*, and *CD36*, and upregulation of *EREG* and *CXCR4*) and modified ncRNAs linked to vascular health, we hypothesized that 1) acute exercise would elicit sex-specific changes in inflammatory and vascular pathways and 2) exercise training would modulate the magnitude and/or direction of the acute exercise-induced monocyte transcriptomic response. Overall, we expected acute exercise to generate a distinct monocyte genomic signature consistent with improved cardiovascular health, while also reflecting transient inflammatory responses associated with acute physiological stress.

METHODS

Study Design

Twelve healthy males and nine females (13–17 yr old) (Supplemental Table S1) completed an 8-wk supervised intervention program primarily of endurance training, with only brief supplementary strength work (1 h/session, 3 days/wk). Sexual maturation was assessed at baseline using Tanner staging and summarized descriptively by sex. $\dot{V}O_{2\text{peak}}$ was assessed before and after the intervention program using a ramp-type progressive cycle ergometer test performed to the limit of the participant's tolerance. On a separate laboratory visit, blood was collected at baseline and immediately after 10, 2-min bouts of cycle ergometer exercise (Fig. 1) interspersed with 1-min rest, a protocol we have used to gauge physiologic and metabolic responses to exercise in youth (19). Elite athletes, individuals who participated vigorously in competitive sports, and potential volunteers with a history of any chronic medical conditions or medication use were excluded from participation. The Institutional Review Board at the University of California, Irvine approved the study, and written informed assent (child) and consent (parent/guardian) were obtained from all participants upon enrollment.

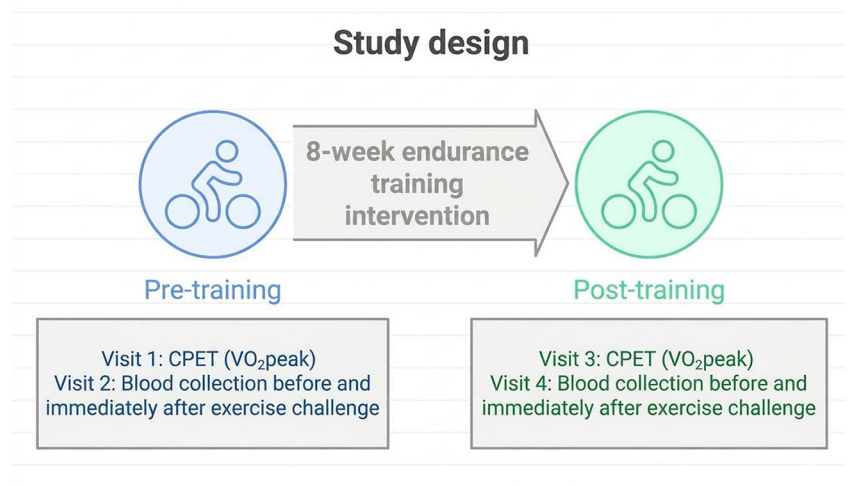
Anthropometric Measurements

Standard calibrated scales and stadiometers were used to determine height and body mass. Body mass index (BMI) was calculated as kg/m^2 and expressed as age- and sex-adjusted percentiles using CDC growth charts.

Measurement of Fitness

Each participant performed a ramp-type progressive cycle ergometer (Ergoline 800 S, Yorba Linda) exercise test using the SensorMedics metabolic system (ENCORE 229 system). After sitting comfortably without pedaling ("resting") on the cycle ergometer for 3 min and 1 min of unloaded pedaling, the work rate (WR) was incremented at 10–30 W/min to the limit of the subject's tolerance. Participants were vigorously encouraged during the high-intensity phases of the exercise protocol. Gas exchange was measured breath-by-breath and

Figure 1. Twelve healthy males and 9 females (13–17 yr) completed an 8-wk endurance exercise intervention program (40–50 min/session, 3 days/wk). $\dot{V}O_{2\text{peak}}$ was assessed before and after the intervention program by a cardiopulmonary exercise test (CPET) (visits 1 and 3). On a separate laboratory visit, blood was collected at baseline and immediately after 10, 2-min bouts of cycle ergometer exercise interspersed with 1-min rest at 70% of the maximum work rate obtained in the CPET (visits 2 and 4).



$\dot{V}O_{2peak}$ was defined as the highest 20-s rolling averaged oxygen uptake achieved during the last 2 min of the test. All participants met standard physiologic criteria for maximal effort, including a respiratory exchange ratio ≥ 1.1 .

Acute Exercise Protocol

At least 48 h but not exceeding 10 days following the completion of the ramp test, each volunteer performed an intermittent-exercise protocol. This exercise regimen was designed to more closely mimic naturally occurring patterns of physical activity in children and has been used extensively over the years in our laboratory. The protocol consists of 10, 2-min bouts of cycle ergometry at 70% of maximum work rate, equivalent to $\sim 75\%$ of $\dot{V}O_{2peak}$, with a 1-min rest interval between each bout. The participants were instructed not to perform any type of vigorous physical activity 48 h prior to the exercise challenge.

Training Program

Participants completed a supervised exercise training program 3 days/wk, consisting primarily of endurance exercise (EE) with brief supplementary strength work. The aerobic component lasted 40–50 min and was performed at 65%–75% of age-predicted maximal heart rate (HR_{max}). Each session also included 5–8 min of muscle-strengthening exercises (M2 multi gym and mat-based exercises), in addition to warm-up and cool-down. The EE intervention was progressive, with intensity increased over time; session frequency and duration were held constant following a 2-wk adaptation period. Sessions used aerobic modalities (e.g., cycle ergometer, treadmill walking/jogging, elliptical, and rowing). Heart rate was monitored continuously (Polar E600) to ensure participants met the prescribed, week-specific target HR.

Blood Sampling and Analysis

Participants arrived at the laboratory between 7:30–8:30 AM after an overnight fast. An indwelling catheter was inserted into the antecubital vein. A baseline sample was taken 30 min after the placement of the catheter and before the onset of exercise. We waited 30 min to ensure that measurable physiological stress parameters (e.g., heart rate and blood pressure) were at baseline levels. The volunteers then completed the intermittent exercise protocol, and additional blood samples were obtained immediately after exercise. Complete blood counts for white blood cell analysis were obtained by standard methods from the clinical hematology laboratory. Plasma lactate was measured before and after exercise using YSI 2300 STAT Plus Glucose & Lactate Analyzer.

Monocyte Isolation

PBMCs were isolated from EDTA-treated blood using OptiPrep Density Gradient Medium (SIGMA). Monocytes were isolated from PBMCs using a negative magnetic cell separation method (Miltenyi Biotec Monocyte Isolation Kit II Human and the autoMACS Pro Separator). The Monocyte Isolation Kit was developed to isolate monocytes from human PBMCs with minimum perturbation. Non-monocytes, i.e., T cells, NK cells, B cells, dendritic cells, and basophils, are indirectly magnetically labeled using a cocktail of biotin-conjugated antibodies against

CD3, CD7, CD16, CD19, CD56, CD123 and Glycophorin A, and Anti-Biotin MicroBeads. Our monocyte isolation procedure consistently yielded high-purity monocytes, with an average purity of 90.8% (ranging from 79% to 97%), as shown by flow cytometry (stained with CD14-FITC).

RNA Extraction

Total RNA for transcriptome analysis (RNA-seq) was extracted from isolated monocytes using TRIzol (Gibco BRL Life Technologies, Rockville, MD). RNA was purified using Qiagen-RNeasy Mini Kit. RNA pellets were resuspended in diethyl pyrocarbonate-treated water. RNA integrity was assessed (prior to beginning target processing) using Agilent Bioanalyzer 2100 (Agilent Technologies, Inc., Palo Alto, CA). We analyzed samples with RNA integrity number (RIN) between 6.7 and 10 (average 9.59).

Library Construction Protocol and Sequencing

The RNA-seq libraries were prepared using the Illumina Stranded mRNA Sample Preparation Kit, including poly(A)-selected mRNA enrichment, fragmentation, cDNA synthesis, adapter ligation, and PCR amplification. Libraries were quantified by KAPA qPCR, multiplexed, and sequenced on an Illumina NovaSeq 6000 platform using paired-end 2 × 100 bp reads at a sequencing depth of ~ 30 million reads/sample.

Bioinformatic Analysis

All participants (21 adolescents) had valid monocyte transcriptome data at four time points, except for one female participant who had a missing blood sample post-acute exercise session conducted after the training period. The quality of the sequencing was first assessed using the fastQC tool (v.0.11.9). Raw reads were quality-trimmed and filtered to a length of 20 bases using trimmomatic (v.0.39). Trimmed reads were mapped to the human GRCh38 reference genome using a splice-aware aligner HiSat2 (v.2.2.1), resulting BAM files were processed using SAMtools (v.1.3), and gene expression was quantified with featureCounts (subread v.2.0.1, featureCounts v.1.5.0-p3) as raw counts. The RNA-seq data are available in Gene Expression Omnibus (GEO) repository, accession number GSE324210. Raw counts were normalized and transformed using limma-voom. Principal component analysis (PCA) identified one male participant with two time points (pre- and post-acute exercise session before the training period) as outliers, and these samples were excluded from downstream statistical analyses. The PCA revealed that biological sex represented a major source of transcriptomic variance in monocytes, with partial separation of female and male samples along the principal components (Supplemental Fig. S1). We conducted linear models for differential expression using the R package limma (v.3.50.3), and a Benjamini–Hochberg adjusted *P* value < 0.1 (false discovery rate [FDR]) was considered statistically significant for each paired comparison; this threshold was selected a priori to balance discovery and type I error in an exploratory transcriptomic data set. In addition, we considered a nominal *P* value < 0.01 to explore different gene expression patterns in response to exercise between males and females (i.e., multifactor design, sex-by-time interaction).

QIAGEN's Ingenuity Pathway Analysis (IPA, QIAGEN Redwood City, www.qiagen.com/ingenuity) (20) was applied to identify IPA canonical pathways altered by exercise, using an absolute activation Z-score threshold >2 and FDR < 0.1 for statistical significance. Activation Z-score calculations have been described previously (20). Differentially expressed genes and ncRNAs (FDR < 0.1) were entered as input, with all quantified transcripts from the study (>23,000) serving as the background data set. In addition, the Database for Annotation, Visualization and Integrated Discovery (DAVID) (21) was utilized to identify exercise-altered molecular pathways via the Kyoto Encyclopedia of Genes and Genomes (KEGG) bioinformatics resource. An EASE score <0.1 was considered statistically significant. The EASE score represents a modified Fisher exact *P* value used for functional enrichment analysis in DAVID, with *P* = 0 indicating perfect enrichment. In silico data mining was conducted using the PHENOPEDIA database to identify genes associated with atherosclerosis and inflammation. PHENOPEDIA curates continuously updated gene-disease associations derived from genetic association studies reported in PubMed (22). Genes altered by exercise in the present study were overlapped with gene lists linked to atherosclerosis and inflammation. Finally, TargetScanHuman 7.2 (https://www.targetscan.org/vert_72/) (23) and LncTarD 2.0 (24) were employed to predict gene targets of exercise-altered microRNAs and lncRNAs.

Statistical Analysis

Student's *t* tests were applied to examine baseline differences in anthropometric and physiological variables. To evaluate training effects between males and females, one-way analyses of covariance (ANCOVA) were conducted for those outcomes (i.e., $\dot{V}O_{2peak}$ and body weight) measured at two time points, with change scores entered as dependent variables, sex (male, female) as the fixed factor, and baseline values included as covariates. For those outcomes (i.e., lactate and monocyte counts) assessed at four time points, two-way repeated-measures analyses of variance (ANOVA) were conducted to determine sex-by-time interaction effects, followed by Bonferroni post hoc testing. Statistical significance was defined as *P* < 0.05. All statistical analyses were carried out using SPSS software version 22.0 (IBM Corporation, Chicago, IL).

RESULTS

Anthropometric and Physiological Characteristics of the Participants

Twelve males and nine females (15 ± 1 yr) with comparable anthropometric characteristics were included in the analysis (*P* > 0.05; Table 1). Baseline Tanner stage is reported descriptively in Table 1. Most participants were in late pubertal development, with Tanner stage 4–5 observed in 67.0% of males and 100% of females. At baseline, before training, males had higher $\dot{V}O_{2peak}$ than females (49.7 ± 8.0 vs. 35.7 ± 3.3 mL·kg⁻¹·min⁻¹; *P* < 0.001; Table 1). The training intervention program was successful, showing an increase in $\dot{V}O_{2peak}$ in both sexes (mean increase: 12% in males and 13% in females; both *P* ≤ 0.023), with no significant between-group difference in the magnitude of change (*P* = 0.088; Fig. 2A and Table 1). Post-training, $\dot{V}O_{2peak}$ remained significantly higher in males compared with females (55.3 ± 7.8 vs. 39.8 ± 4.5 mL·kg⁻¹·min⁻¹; *P* < 0.001; Table 1). Body mass and height did not change significantly following training and did not differ between sexes (all *P* ≥ 0.070).

Lactate: From pre- to post-acute exercise, increased by 319% (males) and 357% (females) before training and by 339% (males) and 295% (females) after training (all *P* < 0.001; Fig. 2B). Monocyte counts increased by 103% (males) and 136% (females) before training, and by 100% (males) and 109% (females) after training (all *P* < 0.001; Fig. 2C). No significant sex-by-time interactions were detected for either lactate or monocyte responses to acute exercise (*P* ≥ 0.592).

Monocyte Transcriptome Profile

Differential transcripts.

Prior to the training program, under resting conditions, 9 transcripts were downregulated and 20 upregulated in males compared with females (FDR < 0.1; Supplemental Table S2). These transcripts were associated with biological processes including chromatin remodeling (*ERCC6L*, *KDM5D*, *KDM6A*, *UTY*), histone H3K27 demethylation and multicellular organism growth (*KDM6A*, *UTY*), as well as noncoding RNAs involved in sex-specific chromosome regulation (e.g., *XIST*, *TSIX*, *TTY10*, *TTY14*). No statistically significant differences were observed in the resting monocyte transcriptomic profiles before versus after the training intervention in either sex (FDR ≥ 0.1; Supplemental Table S3).

Table 1. Anthropometric and physiological characteristics of the 21 participants (12 males and 9 females) before and after training

	Before Training			After Training		
	Males (n = 12)	Females (n = 9)	<i>P</i> Value	Males (n = 12)	Females (n = 9)	<i>P</i> Value
Age, yr	15.2 ± 1.3	15.2 ± 1.5	0.931	15.2 ± 1.3	15.2 ± 1.5	0.931
Tanner stage 1–5 (n, %)	2 (1, 8%) 3 (3, 25%) 4 (6, 50%) 5 (2, 17%)	4 (7, 78%) 5 (2, 22%)	-	-	-	-
Height, cm	172.3 ± 7.9	165.9 ± 8.6	0.093	172.9 ± 7.8	166.1 ± 8.4	0.072
Body mass, kg	60.4 ± 9.7	57.9 ± 12.9	0.622	61.5 ± 9.7	58.3 ± 12.5	0.517
BMI, percentile	46.2 ± 24.0	50.1 ± 27.7	0.731	47.0 ± 23.9	50.8 ± 28.0	0.743
$\dot{V}O_{2peak}$, mL·kg ⁻¹ ·min ⁻¹ *	49.7 ± 8.0	35.7 ± 3.3	<0.001	55.3 ± 7.8	39.8 ± 4.5	<0.001

Values are means ± SD. Tanner stage was assessed at baseline and is reported descriptively by sex. *P* values indicate between-sex comparisons at each time point. Missing $\dot{V}O_{2peak}$ value for one male participant after the training period (*n* = 11). BMI, body mass index; $\dot{V}O_{2peak}$, peak oxygen consumption. *Significant difference between sexes before and after training *P* < 0.05.

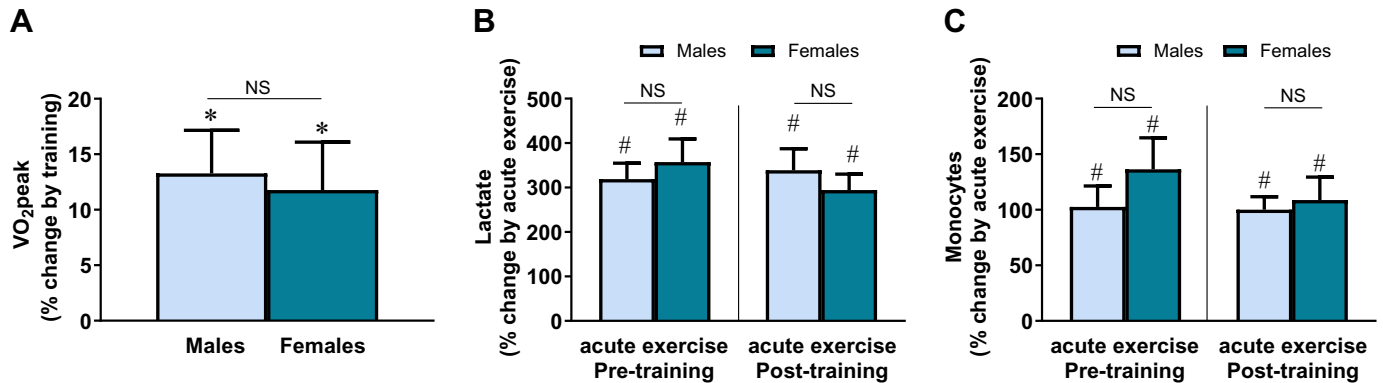


Figure 2. A: $\dot{V}O_{2peak}$ increased significantly after the 8-wk training intervention in both males and females, with no significant sex difference in the magnitude of the training response. B and C: lactate and monocyte counts increased significantly in response to acute exercise in both the males and females, before and after the training intervention. NS, no statistically significant difference between males and females in the response to training or acute exercise. *Significant increase $P < 0.05$ in response to 8 wk of training. #Significant increase $P < 0.05$ in response to an acute bout of exercise. Missing $\dot{V}O_{2peak}$ values for one male participant after the training period ($n = 11$) and monocyte count data for one male participant before the training period ($n = 11$).

Acute endurance exercise, before the training intervention, altered the expression of 567 annotated transcripts in males (FDR < 0.1 ; 184 downregulated and 383 upregulated; Fig. 3A and Supplemental Table S4) and 5,135 in females (FDR < 0.1 ; 2,396 downregulated and 2,739 upregulated; Fig. 3C and Supplemental Table S4). Of the exercise-responsive transcripts identified before training, 314 demonstrated sex-specific expression patterns (sex-by-time interaction, $P < 0.01$; Table 2 and Supplemental Table S5).

After the training period, acute exercise altered the expression of 165 transcripts in males (FDR < 0.1 ; 60 downregulated and 105 upregulated; Fig. 3B and Supplemental Table S4) and 94 in females (FDR < 0.1 ; 47 downregulated and 47 upregulated; Fig. 3D and Supplemental Table S4). Compared with pre-training responses, the acute exercise-induced transcriptomic response was markedly attenuated in both sexes, corresponding to an ~70% reduction in males and a ~98% reduction in females. Among all post-training exercise-responsive transcripts identified in males and females (165 vs. 94, respectively), 26 showed sex-specific expression patterns based on sex-by-time interaction analysis ($P < 0.01$; Table 2 and Supplemental Table S5).

Twenty-three genes and one long noncoding RNA (*SCAMP1-AS1*) were consistently altered by acute exercise across all paired comparisons (both sexes before and after training; Supplemental Table S4).

Molecular pathways.

Heat map shows different activation patterns for IPA canonical pathways altered by acute exercise in monocytes from males and females, before and after training respectively (Fig. 4). Before training, several gene pathways involved in mitochondrial translation, energy production, or protein synthesis were activated by acute exercise in females (Z-score ranged from 4.84 to 6.46; FDR < 0.1) but not in males (FDR > 0.1 and/or absolute Z-scores < 2). Coronavirus pathogenesis and mitochondrial dysfunction pathways were inhibited by acute exercise in females (Z-score -2.14 and -2.55 respectively; FDR < 0.1) and activated in males, although not statistically significant in the latter (FDR > 0.1 and/or absolute Z-scores < 2). Aldosterone signaling in epithelial cells and

HSP90 chaperone cycle for steroid hormone receptors in the presence of ligand pathways, and pyroptosis were activated by exercise in males (Z-score ranged from 2.00 to 2.83; FDR < 0.1) but not in females (FDR > 0.1 and/or absolute Z-scores < 2). After the training period, acute exercise only altered aldosterone signaling in epithelial cells and the HSP90 chaperone cycle for steroid hormone receptors in the presence of ligand pathways in males and females, respectively (FDR < 0.1 ; Supplemental Table S6). Individual genes identified in each IPA canonical pathway, activation Z-scores, and FDR values are provided for each comparison in the Supplemental Table S6.

DAVID pathway enrichment analysis identified 46 and 19 KEGG pathways altered by acute exercise in females and males, respectively, before training (EASE score < 0.1 ; Supplemental Table S7). After the training intervention, no significant pathways were detected in females (EASE score > 0.1), whereas only two inflammatory pathways were identified in males (Herpes simplex virus 1 infection and Legionellosis; EASE score < 0.1 ; Supplemental Table S7).

Before training, six pathways related to viral and infectious processes (e.g., COVID-19, measles, and hepatitis C) were commonly altered in both sexes (Supplemental Table S7). Sex-specific pathway enrichment was also observed: vascular function-related pathways were identified exclusively in males (e.g., Fluid Shear Stress and Atherosclerosis; Lipid and Atherosclerosis), whereas females exhibited enrichment of pathways associated with adipose tissue cross talk and metabolic regulation, including adipocytokine signaling, oxidative phosphorylation, and mammalian target of rapamycin (mTOR) signaling (Table 3).

In silico analyses.

Before training, a proportion of genes altered by acute exercise were associated with atherosclerosis (34 genes in males, 6%; and 300 in females, 6%) or inflammation (69 genes in males, 12%; and 478 in females, 9%); Supplemental Table S8. After training, fewer exercise-responsive genes were linked to these disease categories [atherosclerosis: 10 genes in males (6%) and 4 in females (6%); inflammation: 17 genes in males (10%) and 14 in females (15%); Supplemental Table S8].

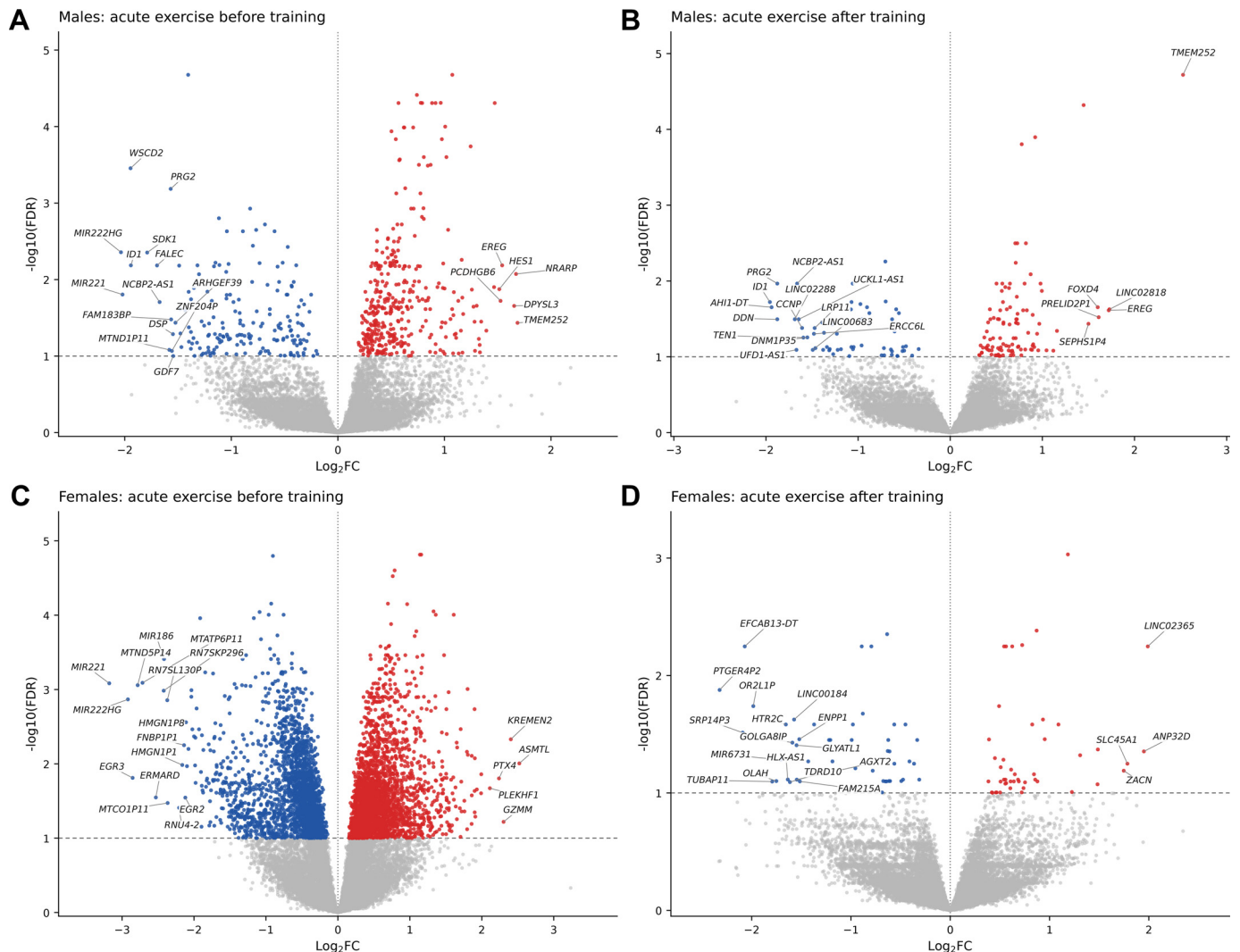


Figure 3. Volcano plot indicating the differentially expressed transcripts (upregulated in red and downregulated in blue) post vs. pre-acute exercise in males (A and B) and females (C and D) before and after the training intervention, respectively. For the analyses examining the effects of acute exercise on monocyte gene expression profiles before training, we included 11 males and 9 females (one male was excluded as an outlier). After training, analyses included 12 males and 8 females due to missing data for one female participant. The x-axis corresponds to the log₂-fold change (FC) value, while the y-axis shows the false discovery rate (FDR). Statistical significance threshold (i.e., FDR < 0.1 corresponds to $-\log_{10} > 1.3$ in the horizontal dashed line). Gene and noncoding RNAs symbols are presented for the 20 transcripts, with the higher magnitude of log₂FC (positive or negative values) in each comparison.

Selected microRNAs and long noncoding RNAs altered by acute exercise with reported cardiovascular relevance were identified based on previous literature (Table 4). Target prediction using TargetScanHuman 7.2 and LncTarD 2.0 indicated that these microRNAs and lncRNAs potentially regulate hundreds of target genes also altered by acute exercise, many of which are associated with atherosclerosis or inflammation according to PHENOPEDIA annotations (Table 4).

DISCUSSION

This study examined the monocyte transcriptomic response to an acute exercise bout before and after an 8-wk endurance training intervention in healthy adolescent males and females. The primary findings are: 1) acute exercise elicited a substantially greater transcriptomic response in females compared with males before training;

and 2) endurance training markedly attenuated the acute exercise-induced response in both sexes. These findings indicate that exercise training modifies immune transcriptional responsiveness during adolescence and reveal sex-specific molecular patterns in pathways relevant to cardiovascular disease. Importantly, the effect of endurance training was evident in the monocyte transcriptomic response to acute exercise, rather than as a detectable change in the resting monocyte transcriptome. Before endurance training, the number of genes affected by brief exercise was approximately ninefold higher in females than in males. Endurance training significantly attenuated the acute exercise response in both sexes; a 98% reduction in females and a ~70% reduction in males relative to pre-training responses.

We carefully adjusted the work rate for each individual (70% of the maximum work rate, equivalent to ~75% of $\dot{V}O_{2peak}$). As shown, key indices of exercise stress, including circulating lactate levels, did not differ between the acute

Table 2. Selected Top transcripts differentially altered by acute exercise between males and females before and after training that are associated with cardiovascular health

Gene or Noncoding RNA	Acute Exercise before Training (log ₂ FC)			Possible Link to Cardiovascular Health
	Males (Post vs. Pre log ₂ FC; Supplemental Table S4)	Females (Post vs. Pre log ₂ FC; Supplemental Table S4)	Interaction (log ₂ FC Direction Males/Females; Supplemental Table S5)	
<i>MIR186</i>	0.21	−2.42	2.63	This microRNA has been considered a diagnostic biomarker in atherosclerosis (25).
<i>ERMARD</i>	−0.14	−2.53	2.38	It has been considered a novel candidate gene for congenital heart defects (26).
<i>KREMEN2</i>	−0.46	2.40	−2.86	Methylation status of this gene was associated with lipidomic profiles from the STANISLAS Family study (27).
<i>TBC1D3F</i>	−1.20	1.59	−2.78	This gene is involved in tissue repair and micropinocytosis (28).
<i>PCDHGA12</i>	−1.08	1.51	−2.60	Deletions within the <i>PCDHA</i> cluster and related genes such as <i>PCDHGA3</i> can contribute to congenital heart disease and ventricular dysfunction (29, 30).
<i>RPL22P3</i>	−1.30	1.16	−2.48	The deletion of ribosomal protein L22 (RPL22) from cells was associated with a higher expression of the C-C motif chemokine ligand 2 (CCL2) following lipopolysaccharide stimulation, promoting the migration of monocytes (31).
<i>IGLL1</i>	−0.51	1.76	−2.29	Genetic variants in this gene were associated with insulin resistance and B-cell dysfunction (32).
Acute exercise after training (log ₂ FC)				
<i>TRDV1</i>	1.38	−1.22	2.62	This gene encodes the TCRδ chain of Vδ1 + γδ T cells, which can modulate atherosclerosis via IL-17 production (33).
<i>MIR99AHG</i>	0.78	−1.55	2.34	This long noncoding RNA improved adipocyte differentiation, and its inhibition prevented obesity in mice (34).
<i>ANP32D</i>	−0.81	1.95	−2.76	<i>ANP32D</i> was upregulated in people with Chronic Mountain Sickness when exposed to low oxygen (35).
<i>USP3-AS1</i>	−1.63	0.76	−2.38	This gene can promote glycolysis and liver metastasis (36).
<i>IFITM10</i>	−1.15	1.17	−2.31	Interferons (IFNs) and IFN-related pathways are involved in cardiovascular diseases (37). Recently, <i>IFITM1</i> was considered a target for atherosclerosis (38). However, there is no specific role for <i>IFITM10</i> .

Selection based on highest absolute log₂ fold changes from interaction analysis; see Supplemental S5. *ANP32D*, acidic nuclear phosphoprotein 32 family member D; *ERMARD*, ER membrane associated RNA degradation; *IFITM10*, interferon induced transmembrane protein 10; *IGLL1*, immunoglobulin lambda like polypeptide 1; *KREMEN2*, Kringle containing transmembrane protein 2; *MIR186*, microRNA 186; *MIR99AHG*, mir-99a-let-7c cluster host gene; *PCDHGA12*, protocadherin gamma subfamily A, 12; *RPL22P3*, ribosomal protein L22 pseudogene 3; *TBC1D3F*, TBC1 domain family member 3F; *TRDV1*, T cell receptor delta variable 1; *USP3-AS1*, USP3 antisense RNA 1.

exercise experiments (Fig. 2B). Consequently, the observed differences in monocyte gene expression responses are unlikely to be explained by differences in exercise dose or physiological stress. The attenuation of the acute monocyte transcriptomic response after endurance training likely reflects adaptive immune remodeling. Repeated exposure to acute exercise over 8 wk may reduce cellular stress and lower transcriptional activation thresholds. Training may also shape monocyte mobilization and subset distribution (18), with different subsets exhibiting unique transcriptomic profiles (65). Previous studies have demonstrated that acute exercise preferentially mobilizes nonclassical monocyte subsets into circulation via catecholamine-dependent β₂-adrenergic signaling (5, 66). Therefore, acute exercise-induced changes in monocyte subset composition may also have contributed to the differential gene expression responses observed in the present study. Because immunophenotypic

characterization of monocyte subsets was not performed, we cannot distinguish whether the observed transcriptomic responses to acute exercise reflect intrinsic transcriptional remodeling within monocytes, shifts in monocyte subset distribution, or a combination of both mechanisms. In addition, our analyses focused on bulk circulating monocytes, which may not capture heterogeneity across monocyte subpopulations or differences between circulating and marginated monocyte pools.

Importantly, given the relatively short lifespan of circulating monocytes, the attenuated acute exercise-induced transcriptomic response observed after endurance training may reflect adaptations beyond the circulating monocyte compartment itself. One possibility is that repeated exercise exposure modifies systemic immune regulation and/or hematopoietic progenitor programming, thereby altering the transcriptional responsiveness of newly generated monocytes to subsequent

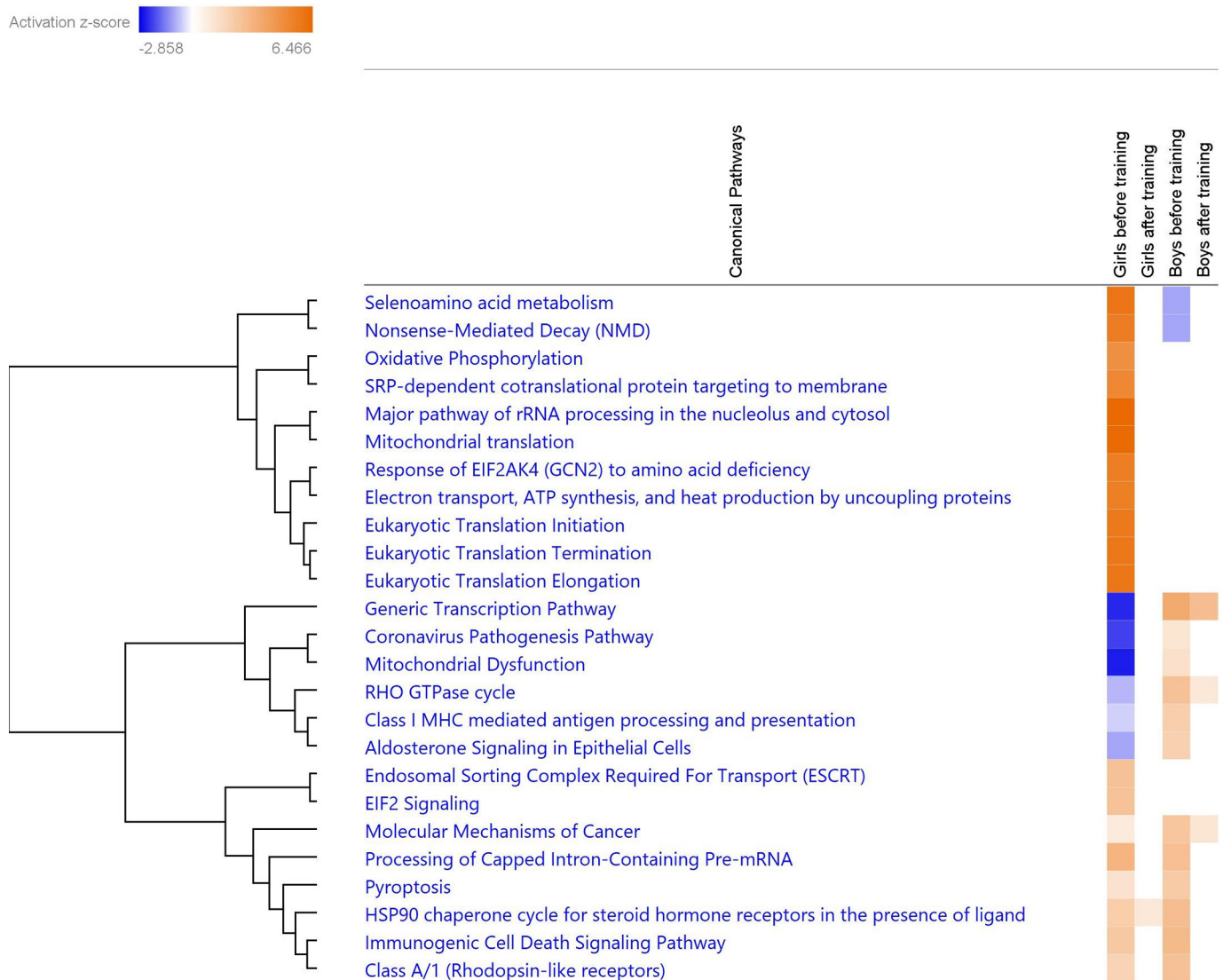


Figure 4. Ingenuity Pathway Analysis (IPA) canonical pathways analysis shows different molecular pathways activation patterns in response to acute exercise before and after the 8-wk training intervention in male and female adolescents.

exercise stimuli. Although speculative, this interpretation is consistent with emerging concepts of exercise-induced immune adaptation and trained innate immunity (67–69). Future studies integrating progenitor-cell, epigenetic, and single-cell approaches will be required to directly test these mechanisms. Together, endurance training may fine-tune the acute transcriptomic responses of circulating monocytes through multiple mechanisms.

At the skeletal muscle and adipose tissue level, it has been demonstrated that exercise training mitigates the metabolic stress and transcriptome responses associated with acute exercise (70, 71). However, little is known about blood components such as circulating monocytes or inflammatory markers. In this regard, in healthy young males, acute high-intensity intermittent exercise induced a smaller increase in serum IL-10 (an anti-inflammatory cytokine) after an 8-wk training intervention compared with the same acute exercise stimulus before training (3). Similarly, our findings indicate that training modulates the acute response to exercise at the

whole-genome level in monocytes from adolescents, markedly reducing the number of transcripts altered by acute exercise in both sexes compared with pre-training responses. Consistent with this pattern, *IL-10* gene expression within monocytes exhibited a lower post-acute exercise increase after training compared with before training in males ($\log_2FC$ 0.72 vs. 1.15) and in females ($\log_2FC$ 1.12 vs. 1.17). The marked attenuation of the acute transcriptomic response after training suggests altered monocyte responsiveness with training, rather than impaired function.

Before training, acute exercise induced a greater monocyte transcriptome response in females compared with males. This finding is consistent with established sex differences in immune regulation, whereby females generally exhibit more robust innate and adaptive immune responses, potentially influenced by sex steroid hormones (72). Following the training intervention, the magnitude of the acute exercise-induced transcriptomic response was reduced in both sexes, resulting in greater similarity of gene expression patterns between

Table 3. Selected gene KEGG pathways (EASE score < 0.1) with implications for cardiovascular health, which were altered by acute exercise before training in males and females, respectively

Category	Term (Number of Genes)	Gene Symbol	Implication for Cardiovascular Health
<i>Acute exercise before training (Males)</i>			
KEGG	Fluid shear stress and atherosclerosis (8)	KLF2, NFE2L2, ARHGEF2, TNFRSF1A, HSP90AB1, PIK3R3, THBD, TP53	Atherosclerosis is the major cause of cardiovascular disease (39).
KEGG	Lipid and atherosclerosis (10)	NFE2L2, NFKBIA, TNFRSF1A, CASP3, CYP2C8, HSP90AB1, HSPA6, HSPA8, PIK3R3, TP53	See above (Term: Fluid shear stress and atherosclerosis) (39).
KEGG	Fc gamma R-mediated phagocytosis (7)	FCGR3A, LIMK2, MYO10, PIP5K1A, PIK3R3, PLA2G4A, PLCG2	The activation of Fc gamma receptors in monocytes/macrophages may contribute to vascular disease pathogenesis (40).
KEGG	Human immunodeficiency virus 1 infection (11)	CCR5, FADD, LIMK2, NFKBIA, TNFRSF1A, APOBEC3G, BTRC, CASP3, CGAS, PIK3R3, PLCG2	This infection can affect vascular biology increasing the risk of cardiovascular disease (41).
<i>Acute exercise before training (Females)</i>			
KEGG	Oxidative phosphorylation (30)	ATP5F1D, ATP5F1E, ATP5MC2, ATP5ME, ATP5MF, ATP6V0A2, ATP6V0E1, ATP6V1A, ATP6V1C1, ATP6V1F, ATP6AP1, NDUF51, NDUF52, NDUF57, NDUF58, NDUFV1, NDUFV2, NDUF11, NDUF12, NDUF13, NDUF14, NDUF15, NDUF16, NDUF17, NDUF18, NDUF19, NDUF20, NDUF21, NDUF22, NDUF23, NDUF24, NDUF25, NDUF26, NDUF27, NDUF28, NDUF29, NDUF30, NDUF31, NDUF32, NDUF33, NDUF34, NDUF35, NDUF36, NDUF37, NDUF38, NDUF39, NDUF40, NDUF41, NDUF42, NDUF43, NDUF44, NDUF45, NDUF46, NDUF47, NDUF48, NDUF49, NDUF50, NDUF51, NDUF52, NDUF53, NDUF54, NDUF55, NDUF56, NDUF57, NDUF58, NDUF59, NDUF60, NDUF61, NDUF62, NDUF63, NDUF64, NDUF65, NDUF66, NDUF67, NDUF68, NDUF69, NDUF70, NDUF71, NDUF72, NDUF73, NDUF74, NDUF75, NDUF76, NDUF77, NDUF78, NDUF79, NDUF80, NDUF81, NDUF82, NDUF83, NDUF84, NDUF85, NDUF86, NDUF87, NDUF88, NDUF89, NDUF90, NDUF91, NDUF92, NDUF93, NDUF94, NDUF95, NDUF96, NDUF97, NDUF98, NDUF99, NDUF100	During diet-induced obesity, inflammatory monocytes require monocytic glycolytic reprogramming with a concomitant decrease in oxidative phosphorylation (42).
KEGG	mTOR signaling pathway (30)	ATP6V1A, ATP6V1C1, ATP6V1F, BRAF, CLIP1, DDIT4, MLST8, NPRL2, NRAS, RICTOR, RAGA, RAGB, SKP2, SOS1, SOS2, TTI1, TNFRSF1A, TSC1, WDR24, WNT5A, WNT5B, CHUK, CASTOR1, DVL1, EIF4B, EIF4EBP1, FNIP1, FZD2, GSK3B, IGF1R, LAMTOR2, LAMTOR4, LPIN2, MAPK1, MAP2K2, PTEN, PIK3CA, PIK3CD, PIK3R1, PRR5L, PRR5, PRKAA1, RPS6KA1, RPS6KA3, RPS6KB2, STK11, SGK1, SESN2, TELO2, ULK1	mTOR regulates inflammatory response in monocytes (43) and can promote atherosclerosis (44).
KEGG	Adipocytokine signaling pathway (45)	CD36, JAK2, NFKBIA, TRAF2, TNFRSF1A, TNFRSF1B, TRADD, ACSL1, ACSL4, APPL1, CPT1A, CHUK, G6PC3, LEPR, MAPK8, PPARA, PCK2, POMC, PRKAA1, PRKAB2, PRKAG2, PTPN11, RXRA, STK11, SOCS3	Dysregulation of adipocytokines contributes to the pathogenesis of various metabolic and cardiovascular disorders (45).
KEGG	FoxO signaling pathway (47)	ARAF, ATM, BRAF, BCL6, EP300, FASLG, KLF2, MDM2, NRAS, RBL2, SKP2, SOS1, SOS2, ATG12, CHUK, CCNB1, CCND1, CDKN1B, FOXO3, FOXO6, G6PC3, HOMER3, IGF1R, IL10, MAPK1, MAPK8, MAP2K2, PTEN, PIK3CA, PIK3CD, PCK2, PIK3R1, PLK2, PRMT1, PRKAA1, PRKAB2, PRKAG2, STK11, STK4, SGK1, S1PR4, TGFBI, TGFBR1	FOXO signaling is activation is related to accelerated atherosclerosis in children with β -thalassemia major (46).
KEGG	Insulin resistance (49)	CD36, CRT2, MLX, NFKBIA, OGA, OGT, TNFRSF1A, CREB1, CREB3, CREB5, CPT1A, G6PC3, GFPT1, GSK3B, MAPK8, NR1H2, NR1H3, PPARA, PTEN, PIK3CA, PIK3CD, PCK2, PIK3R1, PRKAA1, PRKAB2, PRKAG2, PRKCD, PRKCZ, PPP1CA, PPP1CC, PTPN11, RPS6KA1, RPS6KA3, RPS6KB2, SLC27A1, SLC27A4, SREBF1, SOCS3	Insulin resistance is one of the earliest manifestations of a constellation of cardiometabolic diseases that include type 2 diabetes and cardiovascular diseases (47).
<i>Inflammatory pathway altered by acute exercise before training in both males and females</i>			
KEGG	Coronavirus disease—COVID-19 (11 in males and 75 in females)	Males: NFKBIA, TNFRSF1A, C1QB, C5AR1, CGAS, IFNAR2, PIK3R3, PLCG2, RPL21, RPS10, RPS26 Females: OAS1, ADAM17, CXCL8, FAU, FOS, ISG15, JAK1, NFKBIA, TAB2, TRAF6, TNFRSF1A, C1QA, C1QB, C1QC, C5, C5AR1, CHUK, CGAS, CYBB, EIF2AK2, IFNAR1, IFNAR2, IRF3, IL6ST, MAVS, MAPK1, MAPK8, PIK3CA, PIK3CD, PIK3R1, RPL10, RPL10A, RPL12, RPL13, RPL13A, RPL17, RPL18, RPL18A, RPL19, RPL23, RPL27, RPL27A, RPL28, RPL29, RPL3, RPL32, RPL35, RPL36, RPL36A, RPL37, RPL37A, RPL38, RPL7A, RPL8, RPS10, RPS11, RPS12, RPS14, RPS15, RPS16, RPS19, RPS2, RPS20, RPS21, RPS27, RPS29, RPS4X, RPS5, RPS8, RPLP0, RPLP1, RPLP2, TLR2, TLR4, UBA52	COVID-19 viral infection can increase the risk of cardiovascular disease (48).

DAVID, Database for Annotation, Visualization and Integrated Discovery; KEGG, Kyoto Encyclopedia of Genes and Genomes.

Table 4. Selected noncoding RNAs (ncRNAs) involved in cardiovascular health that were significantly altered by acute exercise ($FDR < 0.1$) before and after training in males and females

Acute Exercise before Training (Males)		
ncRNAs (log2FC)	Implication for Cardiovascular Health	Target Genes Affected by Exercise [Using TargetScanHuman 7.2 (23) for microRNA-Genes Interactions and LncTarD 2.0 (24) for lncRNAs-Genes Interactions] and Involved in Inflammation or Atherosclerosis [PHENOPEDIA (22)].
MIR221* (−2.02)	miR-221 expression contributes to coronary artery atherosclerotic plaque (49).	4 target genes: IRF2, SOCS3, COL4A4, CASP10
MIR550A1 (−0.95)	miR-550a mediates the TNF- α function in B cells of patients with diabetes (50).	4 target genes: XCR1, GCH1, PLCG2, CASP10
MIR616* (−1.18)	miR-616 attenuates pulmonary inflammatory response (51).	18 target genes: TNFSF14, CASP3, PRDM1, HSPA6, P2RY2, KMO, MAF, ARHGAP25, CCR5, COL4A4, EREG, ENPP1, HDAC9, MSH6, XCR1, CYP2C8, LIMK2, KLF2
MIR222HG* (−2.04)	Upregulation of this lncRNA is observed in cardiovascular disease (52) and can regulate macrophage polarization (53).	1 microRNA target: MIR221 (disease: pulmonary hypertension)
Acute exercise after training (Males)		
LINC00528 (−0.71)	It has been considered a therapeutic target for myocardial infarction (54).	No targets are affected by exercise.
Acute exercise before training (Females)		
MIR186 (−2.42)	This microRNA has been considered a diagnostic biomarker in atherosclerosis (25).	37 target genes: MAP3K2, NBEAL1, ANO6, SORT1, REL, SCD, KRR1, PTEN, PPIP5K2, PLEK, PTPN11, MAPK1, FAM13A, ATP2B4, DCP1A, AFF1, NFAT5, IPO5, TERF2, CDKN1B, PDS5B, ROCK1, HMGB1, GLIPR1, HPSE, IGF1R, EP300, CYLD, PIK3CA, TNKS2, BACH1, BAZ2B, SMAD2, SPTLC2, MTMR3, MTDH, CADM1
MIR221* (−3.18)	See above (Acute exercise before training; males).	11 target genes: MYLIP, PLCL2, SOCS3, PIK3R1, TERF2, TRPS1, FBN2, COL4A4, NFAT5, NBEAL1, CASP10
MIR222 (−1.69)	miR-221 and miR-222 induces endothelial cell dysfunction promoting the progression of atherosclerosis (55).	14 target genes: CDKN1B, CAMKK1, MDM4, REV1, BAZ2B, ITGB8, GNAI2, ZEB2, ZFH3, DLG2, MAP3K2, CREB1, YOD1, AKAP13
MIR342 (0.83)	This microRNA was proposed as a biomarker for subclinical cardiovascular disease (56).	16 target genes: DTNBP1, MTDH, LRP8, EP300, CLEC7A, DUSP6, MLC1, IPO5, CBX5, RSB1, RPS6KA1, CSK, AFF1, MDM4, PCIF1, SORL1
MIR590 (−1.43)	miR-590 may inhibit atherosclerotic lesion (57).	2 target genes: PPP3CA, HDAC9
MIR616* (−1.91)	See above (Acute exercise before training; males).	139 target genes: TNFSF14, TRIB1, HSD17B7, DLG2, PTPN22, PLCB1, PXX, TPH1, ABL2, IDO1, PTEN, UBE2I, PRDM1, ITGA1, SIRT4, IL6ST, KLF5, PLEKHA1, HSPA6, OLFML2B, CPEB4, VEGFC, FGR, SOD1, RB1, PIK3R1, SEPSECS, RHOB, SPTLC2, P2RY2, JAK1, SNTB1, FEM1B, ZNF300, PRKAG2, GNAQ, NR4A1, KL, UHRF2, IQGAP2, CNR1, NEK4, CBX5, C9orf72, ZNF664, RCN3, PRKAA1, PTPN11, ZEB2, KIDINS220, CD46, PTGER2, ZFH3, OSBPL1A, ANO6, GPR180, PELI1, MDM2, ARHGAP25, CLOCK, RAB31, IGF1R, COL4A4, RSB1, TLR6, CMKLR1, IL13RA1, ACVR2A, AFF1, JAK2, WNT5A, VPS13C, MPDZ, UBE3C, SREBF1, EREG, LTB4R, BACH1, AHR, ADRB1, ENPP1, RCBTB1, DUSP6, CST3, GSK3B, FKBP5, YOD1, ROCK1, HPSE, ATP6V0A2, INHBB, REL, TRAF6, HDAC9, SIRT3, GEN1, GPSM2, GPR132, CD44, CREB1, HMGB1, BTLA, YWHAZ, POT1, TNFAIP8, MAP3K2, ITGB8, SLC9A7, WDFY4, LCAT, NBEAL1, SORT1, RUNX1, MSH6, NDUFS1, TET2, XCR1, H6PD, CLMN, IGF2R, XRCC3, PDS5B, TFAM, SLC11A2, ARHGAP29, BDNF, ESYT2, LGALS1, CADM1, TLR2, ABCA6, SLC19A1, LRP8, APP, BLVRA, ATP2C1, SELPLG, KLF2, FOXO3
MIR629 (−0.84)	This microRNA may regulate pulmonary vascular remodeling (58).	136 target genes: PIN1, DIS3L, ABL2, ATG12, YWHAZ, ZNF300, BSG, KCTD15, ADAM10, ZNF664, CD44, NIN, LST1, RAC2, GLIPR1, NAALADL2, PTPN22, CLDN9, TNFSF14, OLFML2B, MAN2A1, MYC, CASP8, ALCAM, COG5, MBOAT2, VDR, COL4A4, DCP1A, LANCL1, HSPA6, TFAM, APPL1, SNTB1, KCNA3, FBN2, TSC1, CASP5, IFNAR1, PIK3R1, NRIP1, XCR1, RAB31, MAP3K2, ERAP1, SORT1, RHOB, NAT9, B3GNT2, FOXO3, YOD1, TRAF6, AKR1B1, TRPS1, PLCB1, ATP2B4, PGP, KL, WNK1, RUNX3, ZNF366, CTSD, ERCC2, CASP10, PRDM1, IL13RA1, SLC12A6, MMP17, TLR4, BTLA, GPR180, PLCL1, C12orf43, CD93, CBX5, NMNAT2, WNT5A, CMKLR1, ARHGEF2, NFAT5, ATP6V0A2, INPP5D, SEPSECS, GPSM2, SELPLG, AKAP13, SIRT3, ADD2, REST, ROCK1, PRKAB2, SMG6, FOXN3, P2RY2, CIITA, IL16, MSH6, SLC9A7, TGFBRAPI, THBD, ATP2C1, TOLLIP, ZFH3, NBEAL1, IGF2R, FEM1B, NT5C2, PDE7A, CYLD, ACVR2A, MRAS, FKBP5, PPARA, ENTDP1, SOCS3, PRDM2, SPTLC2, NDUFS1, PIK3CA, ITGA1, SMAD2, IPO5, AMACR, SLC27A4, CLMN, TGIF1, YIPF3, IL6ST, CPEB4, IL10RA, GEMIN4, BAZ2B, KLF2, APC, GPR35, ABCA6

Continued

Table 4.— Continued

<i>MIR652</i> (−0.77)	<i>miR-652</i> regulates lipid metabolism and cytokine secretion of macrophages in atherosclerosis (59).	2 target genes: <i>TP53</i> , <i>UBE2I</i>
<i>MIR222HG*</i> (−2.92)	See above (Acute exercise before training; males).	2 microRNAs targets <i>MIR221</i> , <i>MIR222</i> (disease: pulmonary hypertension)
<i>LINC00659</i> (−1.29)	It contributes to deep vein thrombosis by regulating the apoptosis of vascular endothelial cells (60).	No targets are affected by exercise.
<i>APOA1-AS</i> (−0.73)	This lncRNA has been considered a predictive biomarker for atherosclerosis (61).	No results found.
<i>AGAP2-AS1</i> (1.28)	<i>AGAP2-AS1</i> (also known as Punisher) is down-regulated in atherosclerotic plaques (62).	4 target genes: <i>ELAVL1</i> , <i>CFL1</i> , <i>NOTCH2</i> , <i>KLF2</i> (diseases: breast, colorectal and lung cancers).
<i>ATP2B1-AS1</i> (−0.51)	The inhibition of <i>ATP2B1-AS1</i> confers protection against myocardial infarction (63).	No targets are affected by exercise.
<i>HOTAIRM1</i> (0.48)	<i>HOTAIRM1</i> was downregulated in epicardial adipose tissue of atrial fibrillation (64).	<i>ITGA4</i> , <i>PTEN</i> , <i>ZEB2</i> (disease: acute leukemia, gastric cancer, glioblastoma).
<i>Acute exercise after training (Females)</i>		
<i>HOTAIRM1</i> (0.61)	See above (Acute exercise before training; females).	No targets are affected by exercise.

FC, fold change; FDR, false discovery rate. *Noncoding RNAs altered by acute exercise before training in both males and females (FDR < 0.1).

males and females. We also observed a trend for a greater acute exercise-induced monocyte mobilization in females than in males (average increase: 136% vs. 103%) before the training intervention, which was attenuated and more similar in both sexes after the training period (average increase: 109% and 100%) (Fig. 2C). Although not statistically significant, these differences in monocyte mobilization may contribute to the observed transcriptomic patterns and should be considered when interpreting sex-specific responses.

At the pathway level, we observed a significant impact of acute exercise in monocytes only before the training period. Acute exercise activated pathways involved in energy production (e.g., oxidative phosphorylation) and mitochondrial translation while inhibiting coronavirus pathogenesis and mitochondrial dysfunction pathways only in females but not in males (Fig. 4). In general, the activation of oxidative phosphorylation promotes anti-inflammatory programs in immune cells (4), which is linked to enhanced mitochondrial biogenesis and mitochondrial function (4). Also, mitochondrial impairment can contribute to endothelial dysfunction and atherosclerosis (73). At the same time, mitochondrial dysfunction and decreased oxidative phosphorylation were reported in circulating monocytes of patients with COVID-19 pneumonia (74). These observations suggest sex-specific differences in monocyte oxidative and metabolic responses to acute physical stress in healthy adolescents, which may have implications for CVD-relevant immune biology.

Sex-specific differences were observed in cytokine signaling pathways in circulating monocytes of adult patients with coronary artery disease (e.g., chemokine activity genes were downregulated in male patients compared with females) (75). In contrast, these sex-specific differences were not observed in monocytes of healthy adults (75). In our laboratory, we showed that acute exercise altered monocyte gene and microRNA expression profiles to direct monocytes in an anti-inflammatory, antiatherogenic pathway in young males (5). In the current study, we showed that acute exercise (only before training) altered specific pathways involved in vascular function in male adolescents and pathways related to oxidative metabolism and adipose tissue cross talk in female

adolescents (Table 3). These results suggest that endurance exercise training may modulate cardiovascular-related pathways in monocytes of healthy male and female adolescents in a sex-specific manner. Importantly, the identified pathways should primarily be interpreted as mechanistic and hypothesis-generating insights into exercise-induced immune regulation during adolescence, rather than direct biomarkers of clinical benefit.

In previous studies, we highlighted several genes and microRNAs involved in vascular health and/or inflammation (*HDAC9*, *NR4A1*, *VCAN*, *CD36*, *TLR4*, *CXCR4*, *AREG*, *EREG*, *THBD*, *MIR221*) altered by acute exercise in monocytes of young males (22–30 yr old) (1, 5). Similarly, before the training period, we detected these transcripts altered in the same direction (i.e., fold change; Fig. 3 and Supplemental Table S4) by acute exercise in monocytes from female adolescents. Of the aforementioned genes and microRNAs, *HDAC9*, *THBD*, *EREG*, and *MIR221* were altered by acute exercise (preintervention) in both sexes. We observed that *EREG*, which is involved in tissue healing and repair, and vascular smooth muscle growth (1), was upregulated by acute exercise before and after the training intervention only in males, showing one of the highest log₂FC (Fig. 3, A and B). Previously, we reported one of the highest upregulations for *EREG* induced by acute exercise in monocytes of young males (1, 5).

In addition, 23 genes and 1 lncRNA (*SCAMPI-AS1*) were altered by acute exercise independent of sex and training period (Supplemental Digital Table S4). Among these genes, we identified some members of the heat shock protein family (e.g., *DNAJA1*, *HSPH1*), which are involved in atherosclerosis (76) and are affected by acute exercise in monocytes of young males (5). *IRF2BP2*, which has been considered a novel player linking inflammation and lipid metabolism in pediatric obesity (77), and *THBD* [involved in monocyte differentiation in atherosclerotic artery (78)] were also altered by acute exercise in monocytes from young men (5). Our results suggest that these genes could be robust molecular drivers of the immune and vascular response to acute exercise in healthy adolescents.

We identified several ncRNAs altered by acute exercise (before the training intervention) only in males (e.g., *MIR550A1*), females (*MIR186*, *MIR342*, *MIR590*, *MIR652*) or both sexes (e.g., *MIR221*, *MIR616*, *MIR222HG*) with implications for CVD (Table 4), which can bind to genes affected by exercise and involved in atherosclerosis or inflammation based on the PHENOPEDIA database (Table 4). Interestingly, the microRNA *MIR221* and the lncRNA *MIR222HG* (miR222/221 cluster host gene) were the transcripts presenting the highest absolute log₂FC in response to acute exercise (both downregulated) in both males and females before the training intervention (Fig. 3, A and C). The *miR-221/222* cluster is highly expressed in vascular smooth muscle cells and is upregulated in patients with hypertrophic cardiomyopathy (52). *MIR221* upregulation can induce cardiac hypertrophy (in vitro) and heart failure (52), whereas acute exercise downregulated its expression in monocytes of young males (5). Besides, the lncRNA *MIR222HG* has been identified as a key regulator of macrophage polarization (upregulated in M1 macrophages and downregulated in M2 macrophages) and allergic inflammation (53).

Interaction analysis showed that not only genes but also microRNA-related transcripts, long noncoding RNAs, and pseudogenes (i.e., noncoding homologs of protein-coding genes) were among the transcripts exhibiting sex-specific trajectories in response to acute exercise before and after training (Table 2). Some of these genes and noncoding RNAs might be novel modulators of sex-specific immune responses to acute exercise with implications for CVDs (e.g., *IFITM10*, *ERMARD*, *KREMEN2*, *MIR186*, *MIR99AHG*) (Table 2). In recent years, pseudogenes have been considered as relevant transcripts involved in the pathogenesis of CVDs (79). These hypothesis-generating findings may be helpful in directing future research on exercise responses in children and adolescents (80). However, because the libraries were generated using poly(A)-selected RNA-seq rather than specific small RNA sequencing, the study was not specifically optimized for mature microRNA quantification, and miRNA-related findings should therefore be interpreted cautiously.

This study had some limitations. The relatively small sample size is an inherent limitation of mechanistic exercise studies involving transcriptomic analyses in pediatric populations and should be considered when interpreting sex-specific responses. The absence of a nonexercising control group precludes definitive attribution of observed changes to the training intervention alone. Sexual maturation was assessed using Tanner staging, and most participants were in late pubertal development. However, menarcheal status, menstrual cycle phase, and circulating ovarian hormone concentrations were not collected. Therefore, we cannot exclude the possibility that pubertal maturation or hormonal status contributed to the magnitude and variability of the observed monocyte transcriptomic responses. In addition, we did not quantify circulating stress hormones, such as catecholamines and cortisol, and therefore cannot determine whether training-related changes in hormonal responses contributed to the attenuation of exercise-induced transcriptomic responses following the intervention. Finally, transcriptomic analyses were conducted on bulk circulating monocytes, which may not capture heterogeneity across

monocyte subpopulations. Despite these limitations, the within-subject design and integration of acute exercise challenges with transcriptomic profiling provide a framework to examine sex-specific immune responses to exercise during adolescence. Future studies should conduct single-cell RNA-seq (80, 81) to explore the transcriptome response to exercise in specific and novel monocyte subpopulations (65, 82) and to determine the physiological relevance of these transcriptional adaptations in youth, particularly in populations at increased cardiometabolic risk.

Conclusions

Endurance training attenuates the monocyte transcriptomic response to acute exercise in adolescents, with a greater response observed in females prior to training and reduced sex differences following the intervention. This attenuation occurred despite comparable relative exercise intensity and metabolic stress, suggesting that repeated exercise exposure modulates immune transcriptional responsiveness. These findings provide insight into how exercise training influences monocyte behavior during a critical developmental period. The functional implications of these transcriptional changes and their relevance to long-term cardiovascular health remain to be determined.

DATA AVAILABILITY

The RNAseq data are available in the Gene Expression Omnibus (GEO) repository, under accession number GSE324210.

SUPPLEMENTAL MATERIAL

Supplemental Tables S1–S8 and Supplemental Fig. S1: <https://doi.org/10.6084/m9.figshare.31724731>.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

D.M.C. and S.R.-A. conceived and designed research; F.H. and S.R.-A. performed experiments; A.P.-F., F.H., and S.R.-A. analyzed data; A.P.-F., D.M.C., and S.R.-A. interpreted results of experiments; A.P.-F. and F.H. prepared figures; A.P.-F. drafted manuscript; A.P.-F., D.M.C., and S.R.-A. edited and revised manuscript; A.P.-F., F.H., D.M.C., and S.R.-A. approved final version of manuscript.

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